

BI Lab Assignment - 3

Student

student_id

student-name

major

course-section

course_id

section_no

course name

units

room_id

room capacity

Grades

student_id

course_id

prof_id

semester_id

course-grade

(fact table)

Professor

prof_id

prof-name

title

dept_id

dept name

Period

semester_id

year

Q.2

size of the table

- ① student_id : 5 bytes
- ② course_id : 5 bytes
- ③ prof_id : 5 bytes
- ④ semester_id : 5 bytes
- ⑤ course_grade : 5 bytes

Total size per row (Assumption : 25 bytes)

No. of courses : 500

No. of students per course : 60

Time : 30 months

1 sem = 6 months (Assumption)

$$\therefore \text{Total rows} = 500 \times 60 \times 5 = 150000$$

$$\begin{aligned} \text{Size of fact table} &= 150000 \times 25 \\ &= 3750000 \text{ bytes} \end{aligned}$$

course_section

course_id

section_no

course_name

units

room_id

Room

room_id

room capacity

Grades

Student

student_id

course_id

semester_id

course grade

student_id

student_name

major_id

Professor

prof_id

prof name

dept_id

Major

Period

Department

major_id

major name

semester_id

year

dept_id

dept name

course_section

course_id

section_no

course_name

units

room_id

room_capacity

Student

student_id

student_name

major

Attendance

student_id

course_id

semester_id

Period

semester_id

year